

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

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| <b>Course Code:</b>   | <b>CSA12</b>                                                                                                                                                                                                                                              | <b>Course Title:</b> | <b>Computer Architecture</b>                      |
| <b>CO Assessed:</b>   | <b>CO4</b>                                                                                                                                                                                                                                                | <b>Assessment:</b>   | <b>ASSESSMENT TOOL2-<br/>COMPARATIVE ANALYSIS</b> |
| <b>Weightage:</b>     | <b>30%</b>                                                                                                                                                                                                                                                | <b>Total Marks:</b>  | <b>25</b>                                         |
| <b>CO4 Statement:</b> | <i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i> |                      |                                                   |
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| <b>Department:</b>    | B.E.CSE                                                                                                                                                                                                                                                   | <b>Date:</b>         | 22/08/2026                                        |

### ANNEXURE A COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

#### **Code of Conduct:**

*I YAZHINI.E.M 192511165 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**

## RUBRICS FOR CALCULATION:

| Criteria                | Weightage | Level 4 – Exemplary                                                           | Level 3 – Proficient                            | Level 2 – Developing                       | Level 1 – Beginning                      |
|-------------------------|-----------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
| Scope of Items          | 20%       | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%       | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
| Use of Evidence         | 20%       | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided          |
| Critical Thinking       | 20%       | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation; lacks depth          | No meaningful interpretation             |
| Clarity of Presentation | 15%       | Well-organized, clear, and logically structured presentation                  | Generally clear with minor issues               | Somewhat unclear or poorly structured      | Disorganized and difficult to understand |

## **Q1. Compare L1, L2, and L3 Cache Levels — Latency, Bandwidth, Throughput, and Impact on Processor Performance**

Modern processors use a multi-level cache hierarchy (L1, L2, and L3) to bridge the growing speed gap between the CPU core and main memory. Each level trades off capacity, physical proximity to the core, and cost against access speed.

### ***1.1 Latency***

L1 cache is split into instruction and data caches, sits directly beside each core, and returns data in roughly 1 to 4 clock cycles (about 1 nanosecond). L2 cache is larger and slightly farther from the core, adding a latency of about 3 to 10 nanoseconds. L3 cache, usually shared across all cores on the chip, is the largest and physically farthest, with latency in the range of 10 to 30 nanoseconds or more.

### ***1.2 Bandwidth***

Bandwidth generally decreases as cache level increases. L1 offers the highest bandwidth because it uses the widest, shortest on-chip datapaths dedicated to a single core. L2 bandwidth is lower since it is shared between the core's request stream and eviction traffic from L1. L3, being shared across multiple cores through an interconnect or ring bus, has the lowest per-core bandwidth even though its aggregate bandwidth across all cores can be substantial.

### ***1.3 Throughput and Performance Impact***

Throughput — the number of memory operations completed per unit time — closely follows the latency and bandwidth trends. A high L1 hit rate keeps the pipeline fed every cycle, directly maximizing instructions-per-cycle (IPC). When an L1 miss occurs, the penalty of going to L2 (and further to L3 or DRAM on a miss) stalls the pipeline, so the overall Average Memory Access Time (AMAT) is a weighted combination of all three levels' hit rates and latencies. A well-tuned cache hierarchy therefore keeps the vast majority of accesses satisfied by L1 and L2, using L3 mainly as a last-chance buffer before the costly trip to main memory — this is what allows modern processors to sustain high clock speeds without being constantly bottlenecked by memory access.

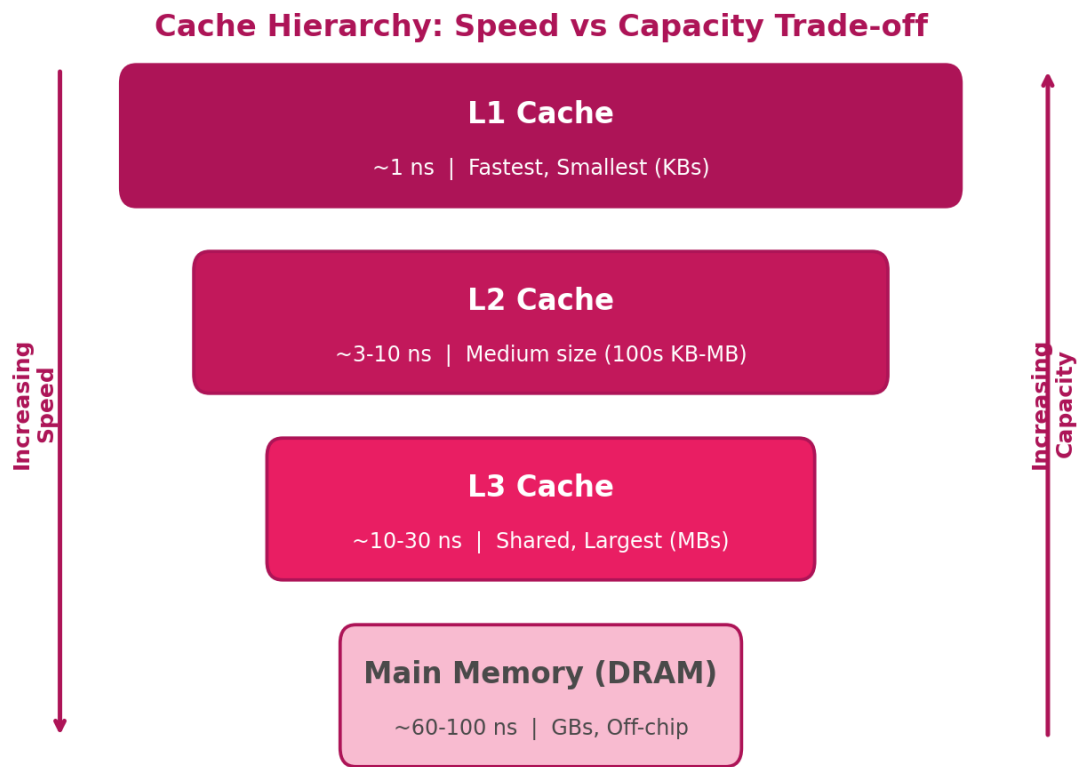


Figure 1: Cache hierarchy showing the trade-off between speed and capacity across L1, L2, L3, and main memory.

| Parameter       | L1 Cache                  | L2 Cache                | L3 Cache                 |
|-----------------|---------------------------|-------------------------|--------------------------|
| Typical Latency | ~1 ns (1-4 cycles)        | ~3-10 ns (10-20 cycles) | ~10-30 ns (30-70 cycles) |
| Typical Size    | 16-64 KB (per core)       | 256 KB-1 MB (per core)  | 2-64 MB (shared)         |
| Bandwidth       | Highest                   | Medium                  | Lowest (per core)        |
| Location        | On-core                   | On-core / near-core     | Shared across cores      |
| Primary Role    | Feed pipeline every cycle | Buffer L1 misses        | Reduce DRAM trips        |

**Q2. Differentiate SRAM and DRAM — Latency, Bandwidth, and Throughput in Cache and Main Memory Design**

SRAM (Static RAM) and DRAM (Dynamic RAM) are the two fundamental memory technologies used in the memory hierarchy — SRAM for on-chip caches and DRAM for main memory.

## 2.1 Structural Difference

SRAM stores each bit using a six-transistor (6T) flip-flop circuit that holds its state as long as power is supplied, requiring no refresh. DRAM stores each bit as a charge on a single capacitor accessed through one transistor (1T1C), which is far denser but leaks charge over time and must be periodically refreshed (typically every 64 milliseconds).

## 2.2 Latency

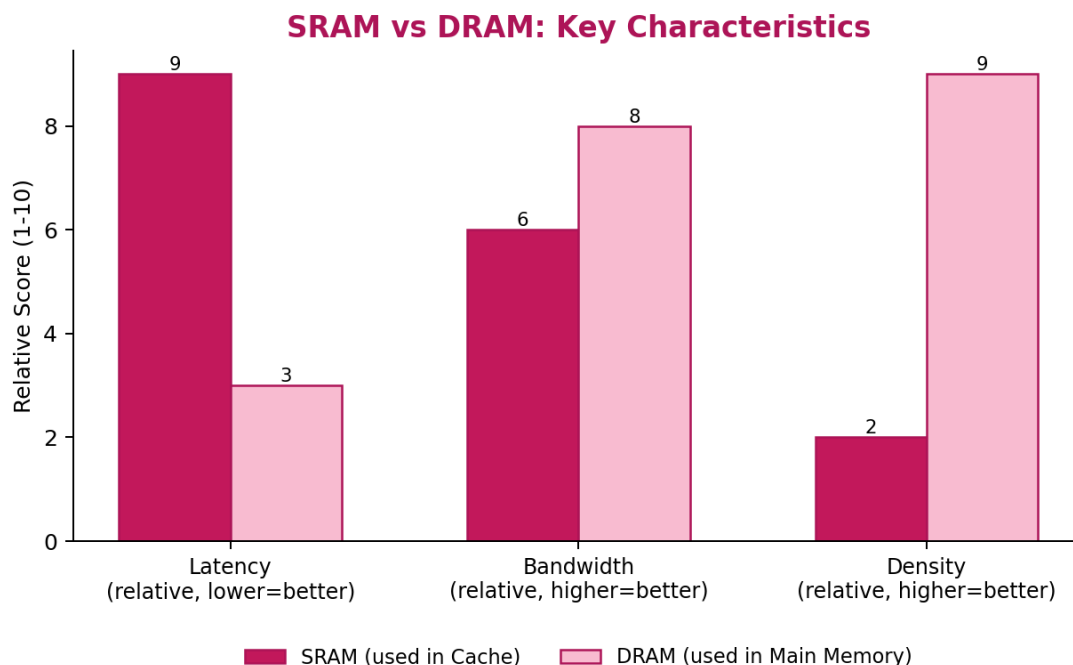
SRAM is significantly faster, with access times in the sub-nanosecond to low single-digit nanosecond range, because reading a stable flip-flop requires no sense-amplification delay comparable to DRAM. DRAM access takes tens of nanoseconds because reading a capacitor's tiny charge requires a slower sense-amplifier cycle, plus a destructive read that must be followed by a write-back to restore the value.

## 2.3 Bandwidth and Throughput

SRAM arrays used in caches can be built with many parallel banks and ports very close to the core, giving high bandwidth for small transfers. DRAM, despite having lower per-access speed, achieves very high aggregate bandwidth through wide buses, burst transfers, multiple banks, and technologies such as DDR/LPDDR and HBM — but its throughput for random, small, latency-sensitive accesses is much lower than SRAM's.

## 2.4 Density, Cost, and Volatility

Because a DRAM cell uses only one transistor and one capacitor versus SRAM's six transistors, DRAM achieves far higher density and lower cost per bit, which is why it is used for gigabyte-scale main memory while SRAM is reserved for kilobyte-to-megabyte-scale caches. Both are volatile, losing their contents when power is removed.



*Figure 2: Relative comparison of SRAM and DRAM across latency, bandwidth, and density.*

| Parameter              | SRAM (Cache)                   | DRAM (Main Memory)                          |
|------------------------|--------------------------------|---------------------------------------------|
| Cell Structure         | 6 transistors (6T), no refresh | 1 transistor + 1 capacitor, needs refresh   |
| Latency                | Very low (~1 ns)               | Higher (~50-100 ns)                         |
| Bandwidth (per access) | High                           | Moderate, but high in aggregate (burst/DDR) |
| Density / Cost per bit | Low density, expensive         | High density, cheap                         |
| Typical Use            | L1/L2/L3 Cache                 | Main Memory (RAM)                           |

### **Q3. Compare Virtual Memory with Paging and Cache Memory Mechanisms — Access Latency and System Throughput**

Both cache memory and virtual memory (implemented through paging) are hierarchical mechanisms that give the illusion of a faster or larger memory than what physically exists, but they operate at very different levels of the memory hierarchy and with very different consequences for latency and throughput.

#### **3.1 Purpose and Mechanism**

Cache memory transparently keeps recently or frequently used data close to the processor in fast SRAM, exploiting temporal and spatial locality; it is managed entirely by hardware using tag comparison and replacement policies (e.g., LRU). Virtual memory, by contrast, uses paging to let a program address more memory than is physically present in DRAM, mapping virtual addresses to physical addresses via page tables maintained by the operating system and accelerated by a Translation Lookaside Buffer (TLB), with pages swapped to and from secondary storage (disk/SSD) on demand.

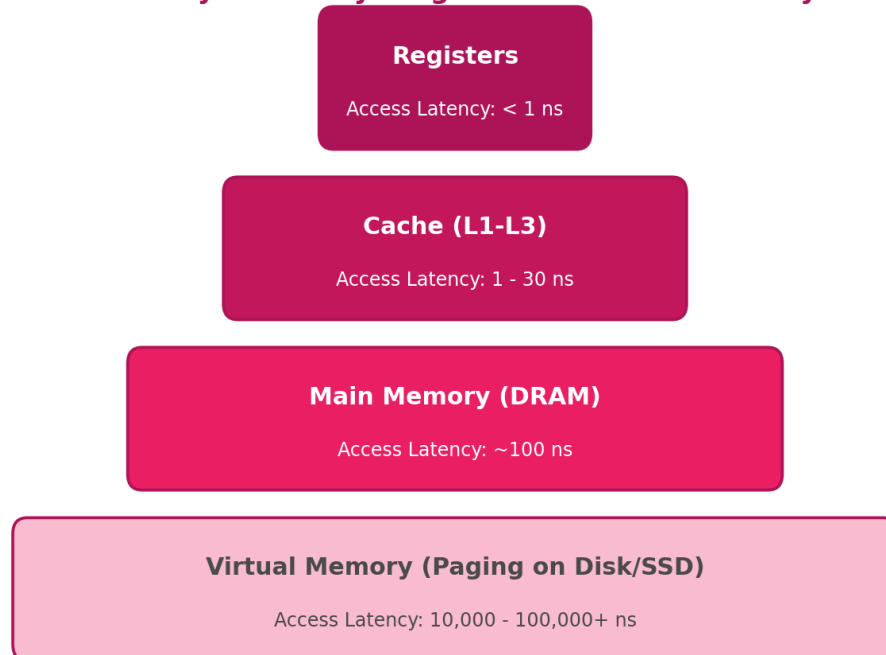
#### **3.2 Access Latency**

A cache hit is resolved in a few nanoseconds, and even a cache miss that must go to DRAM costs only tens to about a hundred nanoseconds. A virtual memory page fault, however, requires an OS trap, a disk or SSD I/O operation, and page-table updates, costing anywhere from tens of microseconds (SSD) to several milliseconds (rotational disk) — a difference of four to six orders of magnitude compared to a cache miss.

#### **3.3 System Throughput Impact**

Because cache misses are relatively cheap and frequent, hardware is designed to tolerate them with techniques like out-of-order execution, prefetching, and multiple outstanding misses, so throughput degrades gracefully. Page faults are rare but catastrophically expensive events for throughput; the OS typically context-switches to another process while the faulting page is fetched, meaning system throughput depends heavily on minimizing the page-fault rate through adequate physical memory and good page-replacement algorithms (e.g., Least Recently Used, Clock).

### Memory Hierarchy: Registers to Virtual Memory



*Figure: The broader memory hierarchy from registers down to virtual memory on disk/SSD, showing the widening latency gap at each level.*

### 3.4 Summary of the Relationship

Conceptually, cache memory and virtual memory are both applications of the same locality-driven hierarchy principle — keep what is likely to be reused close and cheap to access, and push what is not needed immediately farther away — but cache operates transparently in hardware at nanosecond granularity, while virtual memory paging is a software-hardware co-managed mechanism operating at a much coarser, and vastly slower, granularity.

| Parameter          | Cache Memory                                | Virtual Memory (Paging)                                  |
|--------------------|---------------------------------------------|----------------------------------------------------------|
| Managed By         | Hardware (cache controller)                 | OS + hardware (MMU, TLB)                                 |
| Access Unit        | Cache line (e.g., 64 bytes)                 | Page (e.g., 4 KB)                                        |
| Hit Latency        | ~1-30 ns                                    | ~1-2 ns (TLB hit) to ~100 ns (page table walk)           |
| Miss/Fault Penalty | ~60-100 ns (go to DRAM)                     | Microseconds to milliseconds (disk/SSD I/O)              |
| Throughput Effect  | Minor, tolerated via prefetch/OoO execution | Severe; mitigated by sufficient RAM & replacement policy |

#### Q4. Analyze NAND Flash vs DRAM — Latency, Bandwidth, and Suitability for Hierarchical Memory Systems

NAND Flash and DRAM sit at very different points in the storage-memory hierarchy: DRAM as fast, volatile working memory, and NAND Flash as dense, non-volatile secondary storage.

##### 4.1 Latency

DRAM offers access latency in the tens of nanoseconds because reads and writes are electrical operations on capacitor arrays with no mechanical or block-erase overhead. NAND Flash, which stores charge in floating-gate or charge-trap transistors organized into large erase blocks, has read latencies in the tens of microseconds and write/program latencies that can be hundreds of microseconds, with erase operations taking milliseconds — roughly three to five orders of magnitude slower than DRAM.

##### 4.2 Bandwidth

DRAM, connected via a wide, high-frequency memory bus directly to the memory controller, provides bandwidth in the tens of gigabytes per second per channel. NAND Flash bandwidth is constrained by the serial interface to the controller (e.g., via NVMe/PCIe for SSDs) and by internal page/block operation overhead, typically yielding bandwidth in the hundreds of megabytes to a few gigabytes per second — much lower than DRAM, although modern SSDs recover some throughput via heavy internal parallelism across dies and channels.

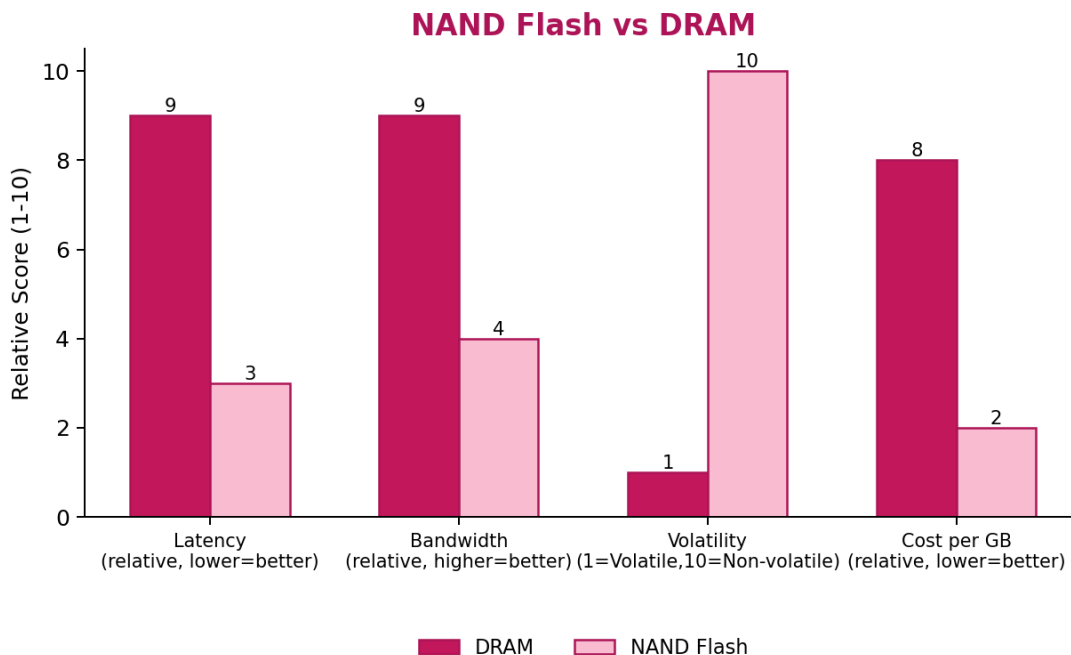


### 4.3 Endurance, Volatility, and Density

DRAM is volatile (loses data without power) but supports effectively unlimited read/write endurance. NAND Flash is non-volatile, retaining data without power, but each cell can only endure a limited number of program/erase cycles before wearing out, requiring wear-leveling. NAND is also significantly denser and cheaper per gigabyte, which is why it is used for bulk, persistent storage rather than as working memory.

### 4.4 Suitability in the Memory Hierarchy

DRAM's low latency and high bandwidth make it ideal as main memory, directly serving the CPU's working set. NAND Flash's non-volatility, density, and lower cost per bit make it ideal as secondary/persistent storage or as a caching layer between DRAM and slower storage (e.g., SSD caching, or as swap space), but its latency is far too high for it to substitute for DRAM in latency-sensitive, byte-addressable roles.



*Figure 3: DRAM vastly outperforms NAND Flash in latency and bandwidth, while NAND wins on non-volatility and cost.*

| Parameter             | DRAM                    | NAND Flash               |
|-----------------------|-------------------------|--------------------------|
| Read Latency          | ~50-100 ns              | ~25-100 microseconds     |
| Write/Program Latency | ~50-100 ns              | ~200-900 microseconds    |
| Bandwidth             | 10s of GB/s per channel | 100s of MB/s to few GB/s |
| Volatility            | Volatile                | Non-volatile             |

| Parameter         | DRAM                      | NAND Flash                                    |
|-------------------|---------------------------|-----------------------------------------------|
| Endurance         | Effectively unlimited     | Limited (P/E cycles); needs wear-leveling     |
| Role in Hierarchy | Main memory (working set) | Persistent/secondary storage, SSD cache layer |

## Q5. Compare MRAM and RRAM — Access Time, Throughput, and Energy Efficiency

MRAM (Magnetoresistive RAM) and RRAM (Resistive RAM) are emerging non-volatile memory (NVM) technologies being explored to close the gap between fast-but-volatile DRAM/SRAM and dense-but-slow NAND Flash.

### 5.1 Operating Principle

MRAM stores data as the magnetic orientation of a free layer in a Magnetic Tunnel Junction (MTJ), read out as a resistance difference (Tunnel Magnetoresistance), and written using spin-transfer torque (STT) or spin-orbit torque in newer variants. RRAM stores data by forming or rupturing a conductive filament within a thin oxide layer sandwiched between two electrodes, switching the cell between high-resistance and low-resistance states.

### 5.2 Access Time and Throughput

STT-MRAM offers very fast access, with read and write latencies in the low nanosecond range (often quoted around 2 to 20 nanoseconds), approaching SRAM/DRAM speeds while remaining non-volatile, giving it high throughput suitable even for cache or DRAM-replacement roles. RRAM is also fast relative to Flash, typically in the range of 10 to 50 nanoseconds for switching, but its throughput can be more variable due to filament-formation variability and the need for forming/set/reset operations with less mature process control than MRAM.

### 5.3 Energy Efficiency

RRAM generally has a simpler cell structure (a single resistive element) and can achieve very low switching energy per bit, along with a manufacturing process that is compatible with CMOS back-end-of-line integration, giving it an edge in energy efficiency for large arrays. MRAM's write operation, especially older field-driven variants, can consume more current to reverse magnetization, though STT-MRAM has substantially reduced this energy cost; MRAM also benefits from essentially no leakage in the standby state, unlike DRAM which needs constant refresh.

### 5.4 Endurance and Maturity

MRAM offers very high write endurance (comparable to DRAM/SRAM, often quoted beyond  $10^{12}$  cycles for STT-MRAM), making it attractive for frequently-written applications. RRAM endurance is typically lower (though still far better than NAND Flash) and can suffer from device-to-device and cycle-to-cycle variability, which remains an active area of research.

5.5 Suitability

MRAM's combination of speed, endurance, and non-volatility makes it attractive as a potential replacement for embedded SRAM/DRAM or as a persistent last-level cache. RRAM's simplicity, scalability, and low energy per switch make it attractive for high-density storage-class memory and for emerging applications such as neuromorphic and in-memory computing.

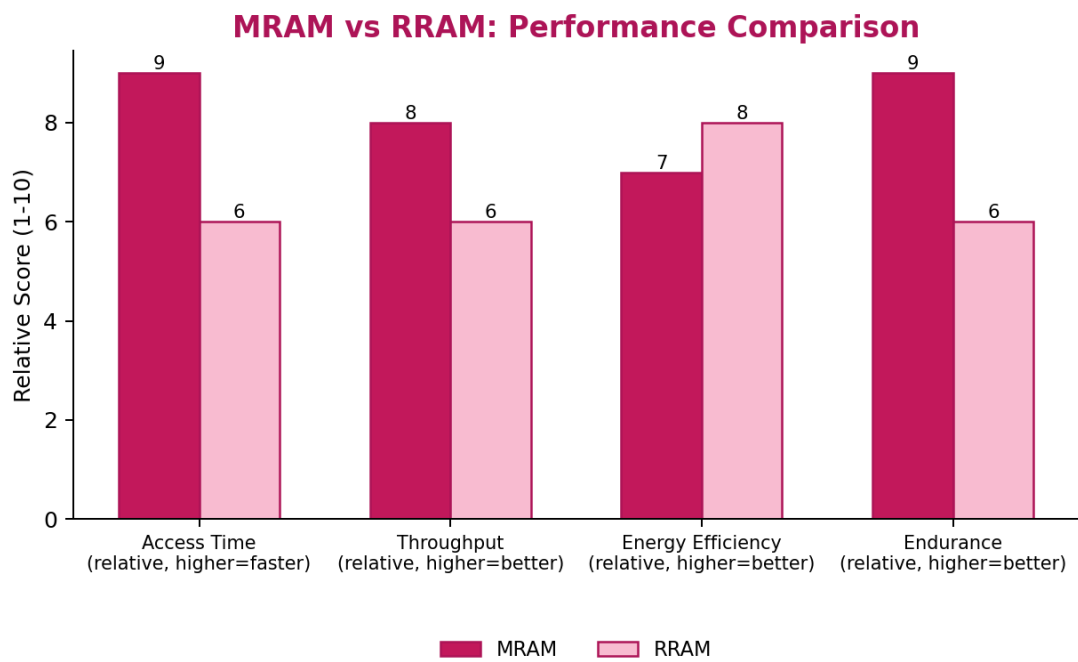


Figure 4: MRAM generally leads in access time, throughput, and endurance; RRAM is competitive in energy efficiency and density.

| Parameter         | MRAM                        | RRAM                                         |
|-------------------|-----------------------------|----------------------------------------------|
| Storage Mechanism | Magnetic orientation (MTJ)  | Conductive filament in oxide layer           |
| Access Time       | ~2-20 ns (STT-MRAM)         | ~10-50 ns                                    |
| Throughput        | High, near SRAM/DRAM levels | Good, some variability from filament forming |

| Parameter         | MRAM                                          | RRAM                                                   |
|-------------------|-----------------------------------------------|--------------------------------------------------------|
| Energy Efficiency | Improved greatly with STT; no standby leakage | Very low switching energy; simple cell                 |
| Endurance         | Very high ( $>10^{12}$ cycles for STT-MRAM)   | Moderate; variability across cycles/devices            |
| Best Suited For   | Cache/DRAM replacement, persistent cache      | Storage-class memory, in-memory/neuromorphic computing |

**Conclusion:** The memory and storage landscape is fundamentally a hierarchy of trade-offs between speed, capacity, cost, and volatility. Traditional levels — L1/L2/L3 cache (SRAM), main memory (DRAM), and secondary storage (NAND Flash) — are increasingly being supplemented by emerging non-volatile technologies such as MRAM and RRAM, which aim to combine the speed of volatile memory with the persistence of Flash, potentially reshaping future memory hierarchies.